Review: Moments of Inertia by Integration

1. Common definition:
$$\boxed{\begin{aligned} I_x &= \int_A \tilde{y}^2\, dA \\ I_y &= \int_A \tilde{x}^2\, dA \end{aligned}}$$

2. But these are only true for a properly chosen dA.
   - Work well for dA parallel to x or y axis.
   - " " " $dA = dx\,dy$ (double integral)

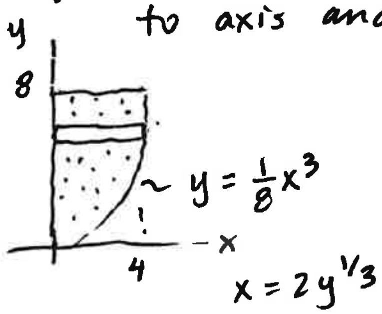
3. If use an element (dA) $\perp$ to axis, must use PAT in differential form:

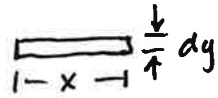
PAT:
$$I_x = \bar{I} + A d_y^2$$
$$I_y = \bar{I} + A d_x^2$$

$$\boxed{\begin{aligned} dI_x &= d\bar{I} + dA\cdot\tilde{y}^2 \\ dI_y &= d\bar{I} + dA\cdot\tilde{x}^2 \end{aligned}}$$

Then, $\boxed{I_x = \int_A dI_x}$ $\boxed{I_y = \int_A dI_y}$

4. Special case of dI_x or dI_y: If dA is $\perp$ to axis and touches the axis:





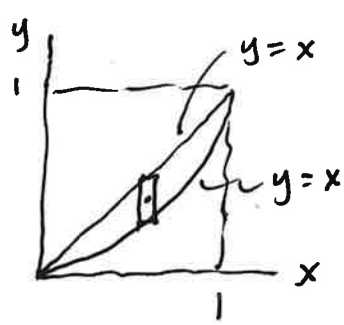
About y axis:

Can use Ibase form..

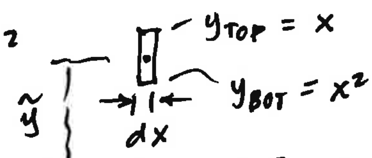
$$\boxed{dI_y = \tfrac{1}{3} b h^3}$$
$$= \tfrac{1}{3}(x^3)\, dy$$

$$I_y = \int_A dI_y = \int_A \tfrac{1}{3} b h^3 = \int \tfrac{1}{3} x^3 dy = \int_0^8 \tfrac{1}{3}(2y^{1/3})^3 dy \quad \text{etc...}$$

5.



If use dA $\perp$ to x (DON'T DO THIS!)



$$\tilde{y} = \tfrac{1}{2}(x^2 + x)$$

$$I_x = \int_A dI_x$$
$$= \int_A \left(\tfrac{1}{12} b h^3 + A\cdot\tilde{y}^2\right)$$
$$= \int_A \tfrac{1}{12}(dx)(x - x^2)^3 + (x - x^2)\,dx\cdot\tilde{y}$$